TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time**: from 09/2026 To 03/2027

(*) Profession: Software Engineer **Specialty**: SE [x]

(*) Kinds of person make registers: Lecturer [x] Students

1. Register information for supervisor (if have)

No.	Full name	Phone	**E-Mail**	**Title**
1	Phạm Minh Trí		tripm14@fpt.edu.vn	Giảng viên
2	Đặng Ngọc Minh Đức		ducdnm2@fpt.edu.vn	Giảng viên

2. Register information for students (if have)

No.	**Full name**	**Student code**	**Phone**	**E-mail**	**Role in Group**
1	Lê Quốc Thống	SE172002	0898243340	Impro2003@gmail.com	Leader
2	Giang Nguyễn Gia Khánh	SE182367	0889779093	Khanhgngse182367@fpt.edu.vn	Member
3	Hà Quang Nghĩa	SE184490	0978874316	Nghiahqse184490@fpt.edu.vn	Member
4	Đặng Đào Anh Đức	SE194009	0376793318	Danganhduc848@gmail.com	Member
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3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

3.1.1. English: ***DroneOps Intelligence – An AI-Assisted Mission Planning, Monitoring, and Incident Analytics Platform for UAV Operations***

3.1.2. Vietnamese: ***DroneOps Intelligence – Nền tảng hỗ trợ lập kế hoạch, giám sát và phân tích sự cố thông minh cho hoạt động UAV***

(*) 3.2. Main proposal content (including result and product)

a) Context:

UAV operations involve mission planning, pre-flight safety checks, telemetry monitoring, and post-flight incident analysis. In many small or experimental drone operations, these activities are fragmented across flight-control software, spreadsheets, maps, and manually reviewed logs.

A mission-management platform can improve traceability and operational safety by validating flight plans before execution, ingesting standardized UAV telemetry, detecting abnormal events, and using historical missions to support risk assessment.

DroneOps Intelligence focuses on the software platform around UAV operations rather than development of the low-level flight controller.

b) Proposed Solutions:

Manage UAVs, pilots/operators, missions, waypoints, mission versions, and flight histories.

Provide pre-flight Geofencing & Flight Path Validation.

Check mission waypoints/segments against configured No-Fly Zones, restricted areas, and altitude limits before approval.

Ingest standardized UAV telemetry using MAVLink, bridged to platform services through WebSocket or MQTT.

Monitor location, altitude, speed, battery, connectivity, and mission progress in real time.

Detect and record operational incidents and abnormal telemetry.

Implement an AI Decision-Support service using similarity retrieval (k-NN/vector search) to retrieve historical missions similar to a proposed/current mission.

Explain risk recommendations using evidence from Top-k similar missions and their incident histories rather than only providing an opaque risk score.

Provide mission dashboards, incident analytics, and post-flight reports.

c) Functional Requirements:

UAV Operator / Pilot:

Register and manage mission plans.

Define routes using waypoints.

Submit missions for pre-flight validation.

View geofence, restricted-zone, and altitude violations.

Monitor UAV telemetry and mission progress.

Receive operational alerts.

Review post-flight mission and incident reports.

Operations Manager:

Manage UAVs, operators, mission approval policies, and operational areas.

Define/import No-Fly Zones, geofences, and altitude restrictions.

Review and approve/reject mission plans.

Monitor active missions.

Review incident histories and risk evidence.

View operational analytics.

Telemetry & Mission Service:

Receive MAVLink-based telemetry through an integration gateway.

Normalize and persist telemetry.

Track UAV and mission state.

Detect communication loss, low battery, abnormal altitude/speed, and configurable operational events.

Maintain mission and incident histories.

Flight Path Validation Service:
Validate waypoints and route segments against geofences/No-Fly Zones.
Validate altitude constraints.
Identify violating route segments and provide evidence on a map.
Prevent approval where mandatory safety constraints are violated.
AI Decision-Support Service:
Construct feature representations for historical and proposed missions.
Retrieve Top-k similar historical missions using k-NN/vector search.
Present similarity, relevant mission characteristics, and historical incidents.

d) Non-Functional Requirements:

Interoperability: MAVLink shall be supported as the UAV telemetry standard at the integration boundary.

Traceability: Mission plan versions, validation results, approvals, telemetry, and incidents must be auditable.

Latency: Real-time telemetry and critical alerts should be delivered with operationally appropriate latency.

Reliability: Temporary telemetry interruptions must not corrupt mission state.

Security: Authentication, authorization, secure API communication, and protected mission data are required.

Explainability: AI decision support must show historical evidence supporting similarity/risk recommendations.

Maintainability: Telemetry adapters, validation rules, and analytics services should be modular.

e) Theory & Practical:

UAV mission management.
Geospatial/geofencing algorithms.
MAVLink telemetry.
Time-series/event processing.
Similarity retrieval and vector search.
Explainable decision-support systems.
Mission planning and map visualization.
MAVLink simulator or compatible UAV/autopilot integration.
WebSocket/MQTT telemetry pipeline.
Spatial database/geospatial operations.
Vector database/search.
Web operations dashboard.

f) Products (Expected Deliverables):

DroneOps Intelligence Web Platform.
UAV Mission Planning and Approval module.
Geofencing & Flight Path Validation Engine.
MAVLink Telemetry Ingestion Gateway.
Real-Time Mission Monitoring Dashboard.

Incident Management and Analytics module.
Similarity-Based AI Decision-Support Service.
Experimental evaluation reports.
Source code, deployment package, and software documentation.

g) Proposed Tasks:

Task Package 1: Requirements, mission workflow, architecture, geospatial data model, and evaluation design.

Task Package 2: Mission planning, geofence management, and flight-path validation.

Task Package 3: MAVLink telemetry integration, mission monitoring, and incident pipeline.

Task Package 4: Historical mission feature engineering, vector search, and similarity-based decision support.

Task Package 5: Integration, evaluation, deployment, testing, and documentation.

3.3. Research Information

a. Research Problem

RQ1: How effectively can automated geofence and flight-path validation detect spatial and altitude constraint violations in UAV mission plans?

RQ2: How accurately and efficiently can the platform ingest and process MAVLink telemetry for real-time mission monitoring and incident detection?

RQ3: Can similarity-based retrieval of historical missions provide relevant and explainable evidence for UAV mission risk assessment?

b. Research Objectives

Develop a software platform for UAV mission planning, monitoring, and incident management.

Develop and evaluate automated flight-path validation.

Integrate MAVLink as a standardized telemetry interface.

Develop and evaluate evidence-based similarity retrieval for mission decision support.

c. Research Scope & Methodology

The project focuses on mission operations software; low-level flight-control algorithm development is outside scope.

Evaluation shall include:

Geofence/path violation detection accuracy.

Telemetry ingestion throughput and latency.

Incident detection performance for defined conditions.

Top-k retrieval relevance using Precision@k/Recall@k or expert relevance assessment.

End-to-end mission workflow demonstrations.

d. Expected Scientific Contribution

An integrated architecture combining UAV mission validation, standardized telemetry, and incident analytics.

An evaluated geospatial flight-path validation approach.

An evidence-based historical mission retrieval method for explainable UAV operational decision support.

4. Other comments (propose all relative things if have):

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HCM, date 05/09/2026

Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*

Automatic flight preparation and workflow ownership

Current ownership: MF1 / WF-01 — NghiaHQ (Ha Quang Nghia), planning and preflight; MF2 / WF-02 — ThongLQ (Le Quoc Thong), realtime monitoring, incidents and flight preparation; MF3 / WF-03 — DucDDA (Dang Dao Anh Duc), post-flight; MF4 / WF-04 — KhanhGNG (Giang Nguyen Gia Khanh), AI assessment. ThongLQ remains team leader and dossier coordinator.

Implemented scope

AutomaticFlightRequested is optional and defaults to false. Changes follow the mission proposal and approval workflow. This release prepares readiness; dispatchEnabled is false. Neither confirmation nor gateway enrollment uploads a mission, arms motors, sends TAKEOFF/AUTO/MISSION_START, or changes the mission to InProgress.

Confirmation conditions

For an approved, opted-in mission with a schedule, the backend opens confirmation at T minus 5 minutes and closes it at the scheduled start. The authenticated operator must keep a visible browser tab online. Presence is sent every 10 seconds and expires after 30 seconds. The selected, unrevoked mission connection must match the active UAV and flight-controller UID, with gateway heartbeat within 30 seconds and telemetry within 15 seconds.

Readiness evidence

Confirmation reruns preflight and binds readiness to the exact plan hash and connection ID. Readiness requires a passing preflight for that hash from the last minute. Changed plan/connection, stale presence or missing telemetry invalidates the conditions; the operator must review and confirm again. Dismissal, a hidden/offline tab or a missed window is not flight consent.

Future execution scope

Autopilot execution remains a separate development phase: firmware/airframe capability detection, mission compilation and upload/read-back, command IDs/deadlines and replay protection, scheduled condition rechecks, local pre-arm/failsafe enforcement, and ACK plus observed flight-state verification. Validate ArduPilot/PX4 SITL, propeller-free bench operation and supervised flight before enabling dispatch. AI advice cannot authorize launch.